

Day 21 — Fri 25 Sep

Countdown: 1 day to General English & GK

Method: start by recalling yesterday's topic with this page closed. Study the sections below, work every example by hand, then do 20 MCQs.

● General — essay technique + models

Source: General/01_GENERAL_PAPER_PLAYBOOK.md

2. ESSAY — 20 marks

The single biggest block on the paper. Fifty minutes, one-fifth of your total.

2.1 How state PSC essays are actually marked

Examiners rarely have a public rubric, but the working marking scheme across state commissions is consistently something like this. Treat it as the shape of the thing, not gospel:

Dimension	Weight	What the examiner is looking for
Content & relevance	~7/20	Do you address <i>this</i> topic? Is there real substance — facts, examples, dimensions — or padding?
Structure & coherence	~5/20	Intro → body with distinct paragraphs → conclusion. One idea per paragraph. Logical flow.
Language & expression	~4/20	Grammar, vocabulary, sentence variety, no repetition.
Balance & maturity	~2/20	Do you see more than one side? Is the tone measured rather than ranting?
Presentation	~2/20	Legibility, margins, paragraph breaks, no overwriting, a visible plan.

The two commonest reasons a competent candidate scores 9 instead of 14:

- They wrote about the *general area* of the topic instead of the *exact* topic. ("Role of technology in governance" answered as an essay on "what is technology".)
- They wrote one enormous 800-word block with no paragraph breaks.

Both are free to fix.

2.2 Length

Target word count	800–1000 words
Absolute floor	600 words (below this it reads as unfinished)
Ceiling	1200 words (beyond this you are stealing time from other sections)
Rough page count	3.5–4.5 sides of standard answer-book paper in normal handwriting
Paragraphs	6–9

Calibrate this **before the exam**: write one timed 900-word essay at home, count the lines, and memorise "900 words = N lines in my handwriting". On exam day you then never count words — you count lines. Saves five minutes.

2.3 The reusable skeletons

You will walk into the hall with **two skeletons memorised**. Whatever the topic, one of them fits.

Skeleton A — the 5-paragraph (use when the topic is narrow or you are short on ideas)

¶	Function	Words
1	Introduction — hook, definition/context, your thesis, roadmap	120
2	The case / the positives / the current situation	180
3	The counter-case / problems / challenges	180
4	The way forward — concrete, specific suggestions	180
5	Conclusion — synthesis + forward-looking close	110
	Total	~770

Skeleton B — the 7-paragraph (use when the topic is broad; this is your default)

¶	Function	Words
1	Introduction — hook / definition / thesis / roadmap	130
2	Background & context — how did we get here? (the <i>past</i>)	130
3	Dimension 1 — usually Economic	140
4	Dimension 2 — usually Social / Human	140
5	Dimension 3 — usually Political/Governance, Technological, or Environmental	140
6	Challenges & the balanced view — "however..."	140
7	Way forward + Conclusion (can be split into two short paragraphs)	150
	Total	~970

Add a Nagaland / North-East paragraph wherever it plausibly fits. On almost any NPSC essay topic — digital governance, education, unemployment, environment, women, tourism, infrastructure — you can insert one paragraph localising the argument to Nagaland or the North-East. Examiners in a state commission notice this and reward it. It is probably the highest-return single habit in this whole document.

2.4 Writing an introduction that scores

A scoring introduction does **four** things in about 120 words, in this order:

1. **Hook** — one arresting sentence. Options:
 - A short relevant quotation.
 - A stark contrast: *"India launches missions to the Moon while a third of its children remain stunted."*
 - A definition: *"Digital governance is the use of information technology to deliver public services more transparently and more cheaply."*
 - A scene: *"In a village in Nagaland's Kiphire district, a mobile signal arrives for two hours a day."* (Use only if plausible — do not invent a specific claim.)
2. **Context** — two sentences placing the topic in time and place.
3. **Thesis** — one sentence stating what *you* will argue. This is the sentence the

examiner looks for. *"This essay argues that X is not primarily a technological problem but an institutional one."*

4. **Roadmap** — one sentence: *"It examines the economic, social and governance dimensions in turn, before turning to the specific case of the North-East."*

Never open with "Since the dawn of civilisation, man has..." or "In this modern era of globalisation...". These are the two most common openings on any answer script and examiners are numb to them.

2.5 Writing a conclusion that scores

Rules:

- **Never introduce a new argument.**
- **Never** just repeat the introduction in different words.
- Do three things in ~110 words:
 1. **Synthesise** — one sentence pulling the body together ("The record is therefore mixed: real gains in access, persistent gaps in quality.").
 2. **Position** — restate your thesis, now as something you have *earned*.
 3. **Forward look** — end optimistic but conditional. "...provided that...", "...if X is matched by Y...". A conditional optimistic close reads as mature; an unqualified "thus India will surely become a superpower" reads as naive.
- A short closing quotation is fine if it genuinely fits. One quotation per essay is plenty; three is showing off.

2.6 Generating content fast under pressure

This is the skill that separates a 9 from a 14 when you don't know much about the topic. Spend your 8 planning minutes running **two frameworks** over the topic and harvesting whatever falls out.

Framework 1 — PESTLE (gives you *dimensions* → your body paragraphs)

Letter	Lens	Ask yourself
P	Political / Governance	Who decides? Which ministry, law, scheme? Centre vs State?
E	Economic	Cost, jobs, growth, inequality, who pays, who profits
S	Social	Families, women, youth, tribes, caste, migration, health, education
T	Technological	What technology enables or threatens this? Digital divide?
L	Legal / Ethical	Which rights are engaged? Article 21? Privacy? Consent?
E	Environmental	Land, forests, rivers, climate, biodiversity, jhum cultivation

Write P-E-S-T-L-E vertically on your rough sheet, spend 4 minutes writing 1–2 bullets next to each, then **pick the best three or four** as your body paragraphs. You now have an essay. You have not needed a single memorised fact.

Framework 2 — Past / Present / Future (gives you *movement*)

Every good essay moves in time.

- **Past**: how did this come about? (¶2 of Skeleton B)
- **Present**: what is the situation now, with its tensions? (body)
- **Future**: what should be done, and by whom? (way forward)

An essay with all three feels complete. An essay stuck in the present feels thin.

Framework 3 — the stakeholder sweep (use when PESTLE runs dry)

List who is affected: **citizen / farmer / student / woman / youth / worker / business / state government / central government / civil society / future generations.**

Each stakeholder is a potential paragraph or at minimum a sentence.

The 8-minute planning ritual (do exactly this)

```
min 0-1  Read all topic options. Pick. Underline the KEY NOUN and the KEY VERB
         in the topic. ("ROLE of technology in RURAL development" – you must
         discuss role, and you must discuss rural.)
min 1-5  PESTLE dump on rough sheet. Past/Present/Future dump.
min 5-7  Choose 5-7 paragraphs. Number them. Write ONE phrase per paragraph
         = your topic sentence.
min 7-8  Write your thesis sentence in full, and your closing line in full.
         (Knowing your last line before you start is a huge stabiliser.)
```

Then write. Do not deviate from the plan even if a better idea arrives at minute 30.

2.7 Using facts, data and quotes

- **A few precise facts beat many vague ones.** Two or three concrete anchors in a 900-word essay is enough. Ten unsupported statistics look invented.
- **Only use numbers you are sure of.** A wrong statistic is worse than no statistic — it signals you are bluffing. If you are unsure, use a *qualitative* claim instead: "a large share of", "the majority of", "roughly half". These cannot be wrong.
- **Safe anchors you can reliably use** (verify the ones you plan to lean on):
- Constitutional articles: Art. 14 equality, Art. 19 freedoms, Art. 21 life and personal liberty, Art. 21A education, Art. 51A fundamental duties, **Art. 371A special provisions for Nagaland**, Art. 243 Panchayati Raj.
- Named schemes and missions (no numbers needed): Digital India, MGNREGA, Ayushman Bharat, PM Jan Dhan Yojana, Swachh Bharat, National Education Policy 2020, Act East Policy, Skill India, Startup India, Jal Jeevan Mission.
- Institutions: NITI Aayog, RBI, Election Commission, CAG, Supreme Court, NEC (North Eastern Council), DoNER Ministry.
- Nagaland-specific: Article 371A; the Nagaland Village Council & Village Development Board system; the Communitisation of Public Institutions and Services Act, 2002 (a genuinely distinctive Nagaland innovation and an excellent essay example); Hornbill Festival; Act East Policy and the state's border with Myanmar.
- **Quotations:** one is enough, two maximum. See [02_ESSAY_BANK.md](#) for a bank of 20. Attribute correctly or attribute vaguely ("as an old proverb has it") — never misattribute to a famous name.

2.8 Common mistakes that lose marks

Mistake	Cost	Fix
No paragraph breaks / one giant block	–3	Break every 120–150 words
Ignoring half the topic ("causes and remedies")	–4	Underline every clause in the question
One-sided rant	–2	Always one "However..." paragraph
Repeating the same idea in ¶3 and ¶6	–2	Follow the plan; each ¶ has one job
Very long sentences (40+ words)	–2	Aim for average 15–20 words; vary length
Overwriting, scratching out, no margin	–2	Cross out with a single clean line

Running to 1500 words and losing time elsewhere	–6 elsewhere	Line-count discipline
Sweeping unverifiable statistics	–2	Qualitative language instead
Slang, SMS abbreviations, "&", "etc." repeatedly	–1	Formal register throughout
Starting with "Since time immemorial..."	–1	Use a real hook

2.9 Handwriting and presentation

Presentation is worth roughly 2 marks and costs nothing.

- **Legibility beats beauty.** If the examiner has to slow down, you lose.
- Leave a **left margin** and a blank line between paragraphs.
- **Underline your essay title.** Underline nothing else except, at most, 2–3 key terms.
An essay with fifteen underlines looks frantic.
- Write the **word count at the end** if you are near target — it signals control.
Skip it if you overshoot.
- Use **black or blue ink only**; one colour throughout.
- If you must delete, one horizontal line. Never scribble a blob.
- **Numbering**: label your answers clearly with the question number. An unlabelled answer can be missed entirely.

2.10 Practice plan for the essay in 21 days

Days 1–3	Read the 4 model essays in 02_ESSAY_BANK.md . Copy out one introduction and one conclusion by hand to internalise the rhythm.
Days 4–14	One full timed essay every alternate day (5 total). 50 minutes, no phone, handwritten. Pick from the topic bank, alternating themes.
Days 4–14 (off days)	10-minute drill: pick a topic, do <i>only</i> the 8-minute plan. No writing. Do 8–10 of these. Plans are cheap and build the reflex.
Days 15–19	2 more timed essays; one must be a Nagaland/North-East topic.
Days 20–21	No new essays. Re-read your own plans and the quotation bank.

● Core — routing, DNS, crypto

Source: [_Shared/CORE_05_ComputerNetworks.md](#)

10. Routing

Concept

Routing is deciding *which path* a packet should take; **forwarding** is the act of moving it out of the right interface. A router keeps a **routing table** whose entries pair a **destination network** with a **next hop**, plus a **metric** (the cost) and often an administrative distance. When a packet arrives, the router finds the **longest matching prefix** in the table and forwards accordingly; if nothing matches, it uses the **default route** (**0.0.0.0/0**), and if there is none, it drops the packet and returns an ICMP "destination unreachable".

	Static routing	Dynamic routing
Configured by	Administrator, manually	Learned automatically by a routing protocol
Adapts to failure	No — needs manual change	Yes — reconverges around a failure
Overhead	None (no protocol traffic, no CPU)	Bandwidth and CPU consumed by updates
Security	More secure/predictable	Updates can be forged unless authenticated
Suits	Small, stable networks ; stub networks	Large, changing networks

Distance Vector vs Link State — the central comparison

Distance vector ("routing by rumour"): each router knows only the **distance** (metric) and **direction** (next hop) to each destination, and it learns them by **periodically sending its entire routing table to its directly connected neighbours**. It never sees the whole topology — it simply trusts what its neighbours tell it. Algorithm: **Bellman–Ford**.

Link state: every router **floods** a small **Link State Advertisement** describing only **its own directly attached links to every router in the area**. Each router therefore builds an **identical complete map** of the topology (the link-state database), and then independently runs **Dijkstra's shortest-path-first algorithm** on that map to compute its own best paths.

Criterion	Distance Vector	Link State
Algorithm	Bellman–Ford	Dijkstra (SPF)
What is shared	The entire routing table	Only link states (LSAs) about own links
Shared with	Directly connected neighbours only	All routers in the area (flooding)
When shared	Periodically (RIP: every 30 s)	Only on change (plus periodic refresh)
Knowledge of topology	Only what neighbours report — no map	Complete map of the area
Convergence	Slow	Fast
CPU / memory	Low	High (must store the whole database and run SPF)
Bandwidth	High periodically (whole table)	Low steady-state (small triggered updates)
Loop problems	Count-to-infinity , routing loops	Essentially loop-free
Loop remedies	Split horizon, poison reverse, hold-down timers, triggered updates, max hop count	Not required
Hierarchy	Flat	Areas (OSPF area 0 = backbone)
Examples	RIP, IGRP	OSPF, IS-IS

The count-to-infinity problem — a favourite 5-marker. If router A's link to network X fails, A hears from neighbour B "I can reach X in 2 hops" (B learnt it from A in the first place). A now believes X is 3 hops away via B; B then hears 3 and says 4, and the metric "counts to infinity" while packets loop. Solutions:

- **Maximum metric** — RIP defines **16 = infinity/unreachable**, capping the count.
- **Split horizon** — never advertise a route back out of the interface you learnt it on.
- **Poison reverse** — actively advertise it back with metric 16 (unreachable).
- **Hold-down timers** — ignore new information about a downed route for a period.
- **Triggered updates** — send changes immediately rather than waiting for the timer.

Routing protocols to know

Protocol	Type	IGP/EGP	Metric	Max hops	Updates	Notes
RIP v1	Distance vector	IGP	Hop count	15 (16 = ∞)	Broadcast every 30 s , whole table	Classful — no VLSM/CIDR support
RIP v2	Distance vector	IGP	Hop count	15	Multicast 224.0.0.9, every 30 s	Classless — carries the mask; supports VLSM; authentication
IGRP / EIGRP	DV / advanced (hybrid)	IGP	Composite: bandwidth, delay, load, reliability	255	EIGRP: triggered, uses DUAL algorithm	Cisco proprietary (EIGRP now open)
OSPF	Link state	IGP	Cost = reference bandwidth / interface bandwidth	Unlimited	Triggered LSAs, multicast 224.0.0.5/6	Open standard , hierarchical areas , area 0 = backbone , fast convergence, VLSM
IS-IS	Link state	IGP	Cost	Unlimited	Triggered	Used in large ISP cores
BGP	Path vector	EGP	AS-PATH and policy attributes	—	Incremental, over TCP port 179	The routing protocol of the Internet ; routes between Autonomous Systems ; chosen for <i>policy</i> , not shortest path

IGP vs EGP: an **Interior Gateway Protocol** (RIP, OSPF, EIGRP, IS-IS) routes **within** a single **Autonomous System**; an **Exterior Gateway Protocol** (BGP) routes **between** autonomous systems. An **AS** is a set of networks under one administrative authority with a common routing policy, identified by an **AS number**.

Other routing concepts named in your syllabus

- **Flooding**: send every incoming packet out of every interface except the one it arrived on. Guarantees delivery and always finds the shortest path, but generates enormous duplicate traffic; controlled by a hop counter or sequence numbers. Used in LSA distribution and military/robust networks.
- **Shortest path routing (Dijkstra)**: build a graph, greedily grow a set of nodes with known shortest distance from the source, relaxing edges as you go.
- **Hierarchical routing**: divide the network into regions; routers know full detail of their own region and only summaries of others. Cuts table size dramatically at a small cost in path optimality — the reason the Internet is routable at all.
- **Broadcast, multicast and anycast routing; spanning tree** for loop-free broadcast.
- **Virtual circuits vs datagrams** — see §12.
- **Concatenated virtual circuits** (explicitly in your Forensic syllabus): when a connection crosses several different networks, a virtual circuit is set up in each network and the

circuits are **spliced/concatenated end to end** at the gateways, so the whole path behaves like one connection-oriented circuit. Contrast with the **connectionless internetworking** approach (what IP actually does), where each datagram is routed independently and may take different routes and arrive out of order. Concatenated VCs preserve ordering and allow resource reservation, but fail entirely if any constituent network fails and cannot route around failures.

Likely exam questions

- **[5]** Differentiate between static and dynamic routing.
- **[5]** Explain the count-to-infinity problem and any two solutions.
- **[5]** What is a routing table? What information does it contain?
- **[5]** Differentiate between RIP and OSPF.
- **[10]** Compare distance vector and link state routing algorithms in detail, naming the underlying algorithm, what is exchanged, with whom, how often, and the convergence and loop characteristics of each.
- **[15]** Explain routing in computer networks: routing tables, static vs dynamic, distance vector (Bellman–Ford) and link state (Dijkstra) with an illustrative topology, the count-to-infinity problem and its remedies, and the roles of RIP, OSPF and BGP.

MCQ traps

Trap	Correct answer
"RIP maximum hop count?"	15 (16 means unreachable).
"OSPF metric?"	Cost (bandwidth-based), not hop count.
"Which algorithm does OSPF use?"	Dijkstra / SPF . RIP uses Bellman–Ford .
"Which is the Internet's routing protocol?"	BGP — a path vector EGP over TCP 179 .
"OSPF backbone area?"	Area 0 .
"RIPv1 supports VLSM?"	No — it is classful. RIPv2 does.
"Split horizon prevents?"	Routing loops / count-to-infinity.
"Distance vector sends what, to whom?"	Whole table , to neighbours only .

15. DNS — Domain Name System

Concept

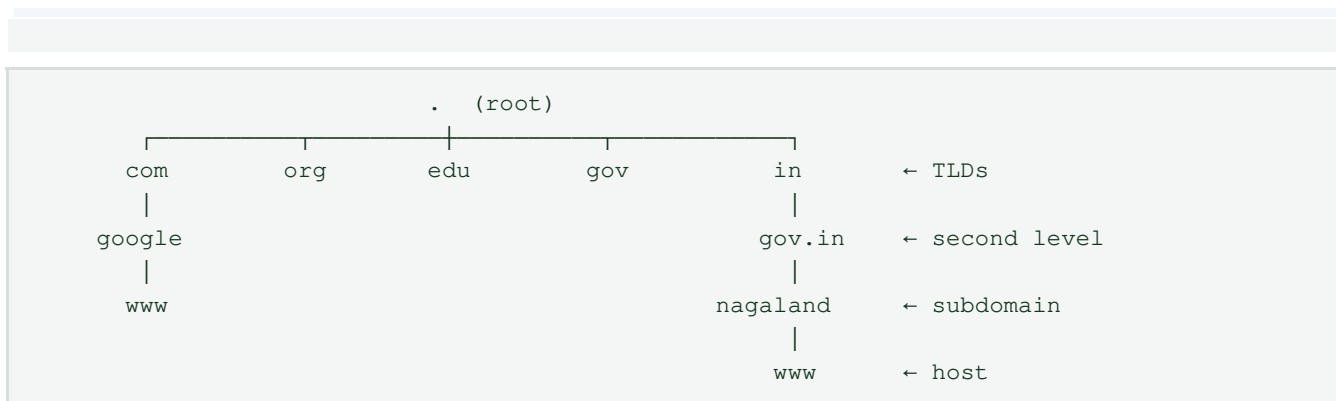
People remember `www.nagaland.gov.in`; routers only understand `104.18.32.7`. **DNS is the distributed database that translates between them.** Two design decisions define it:

1. **It is hierarchical**, so the name space can be delegated. Nobody has to know every name.
2. **It is distributed**, so no single server holds everything and no single failure is fatal.

The name space

An inverted tree again. Read a domain name **right to left**, from most general to most specific. A fully qualified domain name (FQDN) technically ends with a dot for the root:

`www.nagaland.gov.in.`



TLD type	Examples
gTLD (generic)	.com, .org, .net, .edu, .gov, .mil, .int, .info, .biz
ccTLD (country code)	.in, .uk, .us, .jp, .cn
Second level under ccTLD	.co.in, .gov.in, .ac.in, .nic.in

A **zone** is the portion of the name space an individual server is **authoritative** for — the unit of administrative delegation. A **domain** is a subtree of the name space. They coincide unless the domain delegates sub-zones.

Types of name server

Type	Role
Root name server	Top of the hierarchy; knows the authoritative servers for every TLD. There are 13 logical root server addresses (A–M), served by hundreds of physical machines via anycast
TLD server	Authoritative for a top-level domain; knows the servers for each second-level domain
Authoritative server	Holds the actual zone file with the real records for a domain. Primary/master holds the editable copy; secondary/slave copies it via a zone transfer (AXFR/IXFR, over TCP) for redundancy
Recursive resolver / local DNS server	The server your machine is configured to ask (usually your ISP's or 8.8.8.8). It does the legwork on your behalf and caches results
Caching-only server	Holds no zone of its own; just resolves and caches

Resolution — the process to describe

Recursive query: the client asks its local resolver and demands a *final answer* — "either the address or an error, but do not send me elsewhere." The resolver takes on all the work.

Iterative query: the resolver asks a server, which replies "I don't know, but ask *this* server" — a **referral**. The resolver then asks the next one, and so on.

Worked resolution of `www.example.com` (write this out step by step for 10 marks):

1. The browser checks its own cache, then the OS cache, then the local **hosts file** (`/etc/hosts`, or `C:\Windows\System32\drivers\etc\hosts`). Hit → done.
2. Miss → the host sends a **recursive** query to its configured **local DNS resolver**.
3. The resolver checks **its** cache. Miss → it queries a **root** server **iteratively**.
4. The root replies with a referral to the **.com** TLD servers.
5. The resolver queries a **.com** server → referral to the **authoritative** servers for `example.com`.
6. The resolver queries the authoritative server → it returns the **A record**:

`www.example.com → 93.184.216.34.`

7. The resolver **caches** the answer for its **TTL** and returns it to the client.

DNS uses **UDP port 53** for queries (small, fast, retried if lost) and **TCP port 53** for **zone transfers** and responses larger than 512 bytes.

Resource records — explicitly named in your Forensic syllabus

A resource record is a five-tuple: **Domain name, TTL, Class (IN), Type, Value.**

Type	Name	Purpose	Example
A	Address	Maps a name to an IPv4 address	<code>www.example.com. 3600 IN A 93.184.216.34</code>
AAAA	Quad-A	Maps a name to an IPv6 address	<code>IN AAAA 2606:2800:220:1::1</code>
NS	Name Server	Names the authoritative servers for the zone	<code>example.com. IN NS ns1.example.com.</code>
MX	Mail Exchange	The mail server for the domain, with a preference number (lower = higher priority)	<code>example.com. IN MX 10 mail.example.com.</code>
CNAME	Canonical Name	An alias pointing to another name	<code>ftp.example.com. IN CNAME www.example.com.</code>
PTR	Pointer	Reverse lookup — IP → name, in the <code>in-addr.arpa</code> domain	<code>34.216.184.93.in-addr.arpa. IN PTR www.example.com.</code>
SOA	Start of Authority	Zone parameters: primary server, admin email, serial number , refresh, retry, expire, minimum TTL. Exactly one per zone, and it must be first	
TXT	Text	Arbitrary text — used for SPF, DKIM, DMARC (email anti-spoofing) and domain verification	
SRV	Service	Locates the host/port for a named service	<code>_sip._tcp.example.com.</code>
HINFO	Host Info	CPU and OS description (rarely used; an information-disclosure risk)	
CAA	Cert Authority Auth.	Which CAs may issue certificates for the domain	

For the Forensic paper, note the investigative tools: `nslookup`, `dig`, `host`, `whois` (registrant details, registration and expiry dates, name servers), and reverse DNS via PTR. TXT records holding **SPF** and **DKIM** are what you check when analysing whether an email was spoofed.

Likely exam questions

- **[5]** What is DNS? Why is it needed?
- **[5]** Explain the hierarchical structure of the DNS name space with a diagram.
- **[5]** Differentiate between recursive and iterative DNS queries.
- **[5]** What are resource records? Explain A, MX, CNAME, NS and PTR.
- **[10]** Explain the Domain Name System: the name space, zones and domains, types of name server, resource records, and the complete resolution of `www.example.com` step by step.
- **[15]** Discuss DNS in detail — architecture, name servers, resource records, caching and

TTL, recursive vs iterative resolution — and explain DNS-based attacks (cache poisoning, spoofing, tunnelling) and their countermeasures.

MCQ traps

Trap	Correct answer
"DNS port and transport?"	53 , UDP for queries, TCP for zone transfers.
"Record for IPv6?"	AAAA (A is IPv4).
"Record for a mail server?"	MX . For an alias → CNAME .
"Reverse lookup record?"	PTR , under <code>in-addr.arpa</code> .
"How many SOA records per zone?"	Exactly one .
"Lower MX preference means?"	Higher priority.
"Number of root server addresses?"	13 (A–M).
"Which query type does the client send to its resolver?"	Recursive .

18. Network security, cryptography and firewalls

Concept

Security is usually framed as the **CIA triad** plus two:

Goal	Meaning	Threatened by
Confidentiality	Only authorised parties can read the data	Eavesdropping, sniffing, traffic analysis
Integrity	Data is not altered in transit or storage	Modification, replay, MITM
Availability	Services are usable when needed	DoS / DDoS , ransomware
Authentication	Parties are who they claim to be	Spoofing, masquerade
Non-repudiation	A sender cannot later deny having sent	— (provided by digital signatures)

Attack classification — a reliable 5-marker:

	Passive attacks	Active attacks
Nature	Observe only; do not alter data	Alter data or disrupt service
Detectability	Very hard to detect (nothing changes)	Detectable
Prevention	Prevention is the goal (encryption)	Detection and recovery is the goal
Examples	Eavesdropping/sniffing , traffic analysis	Masquerade/spoofing , replay , modification , denial of service , MITM

Named threats to define in one line each:

Threat	Definition
--------	------------

Packet sniffing	Capturing traffic off the wire/air with a NIC in promiscuous mode (Wireshark, tcpdump). Trivial on a hub or Wi-Fi; needs ARP spoofing or port mirroring on a switch
Spoofing	Forging a source identity — IP spoofing, MAC spoofing, ARP spoofing, DNS spoofing/cache poisoning, email spoofing
Man-in-the-middle	Attacker relays and possibly alters traffic between two parties who believe they talk directly
Replay	Capturing a valid message and re-sending it later. Countered by nonces, timestamps, sequence numbers
DoS / DDoS	Exhausting a resource so legitimate users are denied service; distributed when launched from a botnet . Examples: SYN flood, ping of death, smurf, amplification
Phishing	Fraudulent messages inducing the victim to reveal credentials; spear phishing is targeted
Malware	Virus (attaches to a host file, needs a host and user action), worm (self-replicating, self-propagating over a network , needs no host file), Trojan horse (disguised as legitimate, does not self-replicate), ransomware, spyware, keylogger, rootkit, botnet, logic bomb
SQL injection	Injecting SQL through an input field (see the DBMS file)
XSS	Injecting script into a web page viewed by others

Virus vs worm is asked constantly: a **virus requires a host file and human action to spread; a worm is standalone and spreads by itself across a network.**

Cryptography

Encryption turns **plaintext** into **ciphertext** using an **algorithm (cipher)** and a **key**; **decryption** reverses it. **Kerckhoffs's principle**: the security must rest entirely in the **key**, never in the secrecy of the algorithm.

Symmetric (secret key / private key / conventional)

One key is used for both encryption and decryption, and both parties must have it.

- Fast — suitable for **bulk data**.
- **Two problems**: (1) **key distribution** — how do you get the key to the other party securely in the first place? (2) **key explosion** — **$n(n-1)/2$ keys** are needed for n parties to communicate pairwise. For $n = 100$ that is **4,950** keys.
- **Cannot provide non-repudiation** — since both parties hold the same key, either could have produced any message.
- Algorithms: **DES** (56-bit key, 64-bit block — **broken by brute force**), **3DES/Triple DES** (168/112-bit effective), **AES** (Rijndael; **128/192/256-bit keys**, 128-bit block — the current standard), **Blowfish, Twofish, IDEA, RC4** (stream, deprecated), **RC5**.
- Types: **block ciphers** (fixed-size blocks, modes ECB/CBC/CFB/OFB/CTR) and **stream ciphers** (bit/byte at a time).

Asymmetric (public key)

A mathematically related key pair: a **public key**, published to the world, and a **private key**, never disclosed. What one key encrypts, only the other can decrypt. Introduced by **Diffie and Hellman (1976)**.

The two modes — and getting these the right way round is the whole topic:

Goal	Encrypt with	Decrypt with	Why
Confidentiality	The receiver's PUBLIC key	The receiver's private key	Only the receiver holds the private key, so only the receiver can read it
Authentication / digital signature	The sender's PRIVATE key	The sender's public key	Only the sender could have produced it, so it proves origin and gives non-repudiation

Algorithms: **RSA** (Rivest–Shamir–Adleman, 1977; security rests on the difficulty of **factoring large primes**), **Diffie–Hellman** (key *exchange* only, not encryption; based on the **discrete logarithm** problem), **ECC** (elliptic curve — equivalent strength with much shorter keys), **DSA/EIGamal**.

Comparison table:

Criterion	Symmetric	Asymmetric
Keys	One shared secret	Key pair (public + private)
Speed	Fast (100–1000× faster)	Slow
Key distribution	The hard problem	Solved — publish the public key
Keys for n users	$n(n-1)/2$	$2n$
Suitable for	Bulk data encryption	Key exchange, digital signatures, small data
Non-repudiation	No	Yes
Key length	128–256 bits	2048–4096 bits (RSA) for comparable strength
Examples	DES, 3DES, AES , Blowfish, RC4	RSA , Diffie–Hellman, ECC, DSA

The hybrid system — how TLS/HTTPS and PGP actually work, and a superb 10-mark answer: use **asymmetric cryptography** to exchange a randomly generated symmetric "**session key**", then use **symmetric cryptography** for the actual data. You get the key-distribution solution of public key cryptography *and* the speed of symmetric cryptography. Say this explicitly — it shows you understand why both exist.

Hash functions and digital signatures

A **cryptographic hash function** takes input of any length and produces a **fixed-length digest**. Properties: **one-way** (cannot invert), **deterministic**, **avalanche effect** (one bit change → completely different digest), **collision-resistant**.

Algorithm	Digest size	Status
MD5	128 bits (32 hex chars)	Broken — collisions are easy. Still used for file identification in forensics
SHA-1	160 bits (40 hex chars)	Broken (2017) — deprecated
SHA-256 / SHA-512	256 / 512 bits	Current standard
SHA-3	Variable	Newest standard

Hashes give **integrity**, not confidentiality. In forensics, the **hash value of an image is computed before and after acquisition** and compared; matching hashes prove the evidence was

not altered.

Digital signature — the mechanism (draw this):



If **digest₁ = digest₂**, then (a) the message was not altered — **integrity**; (b) it came from the holder of the private key — **authentication**; and (c) the sender cannot deny it — **non-repudiation**.

Why hash the message first instead of signing it whole? Because asymmetric encryption is slow and the message may be huge; hashing reduces it to a fixed small digest. State this.

Digital signature vs encryption:

	Digital signature	Encryption
Provides	Authentication, integrity, non-repudiation	Confidentiality
Key used to create	Sender's private key	Receiver's public key
Key used to verify/open	Sender's public key	Receiver's private key
Is the message hidden?	No — signing does not conceal	Yes

Digital certificate and PKI: a public key alone proves nothing — how do you know it belongs to your bank and not to an attacker? A **Certificate Authority (CA)** issues a **digital certificate (X.509)** binding an identity to a public key and **signs it with the CA's own private key**. Your browser trusts a set of root CAs, so it can verify the chain. That whole apparatus — CAs, certificates, registration authorities, revocation lists (CRL/OCSP) — is the **Public Key Infrastructure (PKI)**.

Firewalls

A **firewall** is a hardware or software barrier placed between a trusted internal network and an untrusted external one, which **inspects traffic and permits or denies it according to a rule set (ACL)**. It enforces the network's security policy at a single choke point.

Type	Layer	How it works	Strengths	Weaknesses
Packet filter	3–4	Examines each packet's source/destination IP, port, protocol and flags against static rules; stateless	Fast , cheap, transparent	No context — cannot tell a reply from an unsolicited packet; cannot see payloads; vulnerable to spoofing and fragmentation attacks
Stateful inspection	3–4	Maintains a state table of active connections ; permits inbound packets only if they belong to a connection initiated inside	Much stronger than static filtering; still fast	Cannot inspect application content; state table is a DoS target

Application-level gateway / proxy	7	A proxy terminates the connection and re-originates it, understanding the specific protocol (HTTP, FTP) and inspecting content	Deep inspection , logging, content filtering, hides internal hosts	Slow ; a separate proxy is needed per application
Circuit-level gateway	5	Validates the TCP handshake, then relays bytes without inspecting them (SOCKS)	Low overhead, hides internal addresses	No content inspection
NGFW / UDTM	3–7	Combines stateful inspection with IPS, deep packet inspection, application awareness, TLS inspection	Comprehensive	Costly, complex, can be a bottleneck

DMZ (Demilitarised Zone) — a subnet between two firewalls (or on a third interface of one) holding **public-facing servers** (web, mail, DNS). If a DMZ server is compromised, the attacker is still separated from the internal LAN by another firewall. Draw this: Internet → outer firewall → DMZ (web/mail servers) → inner firewall → internal LAN.

What a firewall cannot do — worth two marks: it cannot stop attacks that **bypass** it (rogue modems, USB drives, personal hotspots), it cannot stop **insider** attacks, it cannot detect malware in **encrypted** traffic without TLS interception, and it cannot patch a vulnerable application. Complement with **IDS/IPS**, **antivirus** and patching.

IDS vs IPS: an **Intrusion Detection System** monitors and **alerts** (passive, out of band); an **Intrusion Prevention System** sits **in line** and **blocks**. Detection methods: **signature-based** (matches known patterns; misses zero-days) and **anomaly/behaviour-based** (flags deviations from a baseline; more false positives).

VPN: a **Virtual Private Network** creates an **encrypted tunnel** across a public network, so a remote user or branch office behaves as if directly attached to the private LAN. Protocols: **IPSec** (**AH** for authentication/integrity, **ESP** for encryption; **transport mode** protects the payload, **tunnel mode** encapsulates the entire original packet in a new one), **SSL/TLS VPN**, PPTP, L2TP, WireGuard.

Tunnelling, fragmentation and internetworking

These three are named explicitly in your Forensic syllabus, so they need clean definitions.

Tunnelling is **encapsulating one protocol's packet inside another protocol's packet** so that it can traverse a network that does not support it. The classic example: two IPv6 islands separated by an IPv4-only Internet — each IPv6 packet is placed inside an IPv4 packet at the entry router, carried across as ordinary IPv4 payload, and unwrapped at the exit router. The intermediate network never knows IPv6 exists. Applications: **6to4 / Teredo** (IPv6 over IPv4), **VPNs** (IPSec tunnel mode, L2TP, PPTP, GRE), **MPLS**, connecting two remote LANs of the same type over a WAN of a different type. Analogy that works well in an answer: putting a car on a train to cross a mountain — the car doesn't drive through, it is *carried*.

Fragmentation happens because different networks have different **MTUs** (Maximum Transmission Unit — the largest payload a link can carry; **Ethernet's MTU is 1500 bytes**). When a router must forward a packet onto a link with a smaller MTU, it **breaks the packet into fragments**, each with its own IP header.

- IPv4 header fields used: **Identification** (all fragments of one packet share it), **Flags** — **DF** (Don't Fragment) and **MF** (More Fragments; set on all but the last), and **Fragment Offset** (position of this fragment's data, **measured in 8-byte units**).

- **Reassembly is done only at the final destination**, never at intermediate routers, because fragments may take different routes.
- **Costs**: more headers (overhead), more processing, and **loss of any one fragment forces retransmission of the whole original packet**.
- **Transparent fragmentation** — the fragments are reassembled at the exit gateway of the small-MTU network, so the next network sees the original packet.
Non-transparent fragmentation — fragments travel independently all the way to the destination host, which reassembles. **IP uses non-transparent fragmentation**.
- **IPv6 abolishes router fragmentation**: only the **source** may fragment, and it uses **Path MTU Discovery** (send with DF set; a router that cannot forward returns ICMP "fragmentation needed", revealing the path MTU).
- **Forensic relevance**: overlapping and tiny fragments are a classic technique for **evading IDS/firewalls** (teardrop attack, fragment overlap).

Worked fragmentation example. A 4000-byte IP packet (20-byte header + 3980 bytes of data) must cross a link with **MTU 1500**.

- Each fragment may carry at most $1500 - 20 = 1480$ bytes of data, and the data length of every fragment except the last must be a **multiple of 8**. 1480 is a multiple of 8 ✓
- Fragment 1: data bytes 0–1479, **offset = 0**, MF = 1, total length 1500
- Fragment 2: data bytes 1480–2959, **offset = 1480/8 = 185**, MF = 1, total length 1500
- Fragment 3: data bytes 2960–3979 (1020 bytes), **offset = 2960/8 = 370**, **MF = 0**, total length 1040
- Total: **3 fragments**; overhead rose from 20 bytes of header to 60.

Likely exam questions

- **[5]** Differentiate between active and passive attacks with examples.
- **[5]** Differentiate between symmetric and asymmetric cryptography.
- **[5]** What is a digital signature? How does it provide non-repudiation?
- **[5]** What is a firewall? Explain any two types.
- **[5]** Differentiate between a virus, a worm and a Trojan horse.
- **[5]** What is tunnelling? Give one application.
- **[5]** Explain fragmentation. What is an MTU?
- **[10]** Explain public key cryptography. How is it used for confidentiality and for authentication? Describe the working of RSA and of digital signatures with a diagram.
- **[10]** Explain firewalls: types, working, placement, the DMZ, and their limitations.
- **[15]** Discuss network security in detail: security goals, active and passive attacks, symmetric and asymmetric cryptography with a comparison, hash functions, digital signatures and certificates, firewalls and VPNs.

MCQ traps

Trap	Correct answer
"For confidentiality, encrypt with whose key?"	The receiver's public key.
"For a digital signature, encrypt with whose key?"	The sender's private key.
"Which provides non-repudiation?"	Asymmetric / digital signature , never symmetric.
"Keys needed for n users, symmetric?"	$n(n-1)/2$. Asymmetric = $2n$.

"DES key size?"	56 bits (64 with parity). AES = 128/192/256.
"MD5 / SHA-1 digest size?"	128 / 160 bits.
"RSA's hard problem?"	Factoring large numbers . Diffie–Hellman → discrete log .
"Diffie–Hellman encrypts data?"	No — it is key exchange only.
"Does signing hide the message?"	No .
"Which firewall type inspects content?"	Application-level gateway (proxy) , layer 7.
"Ethernet MTU?"	1500 bytes .
"Fragment offset is measured in?"	8-byte units .
"Who reassembles IP fragments?"	The final destination only.
"Virus vs worm?"	A worm needs no host file and self-propagates .